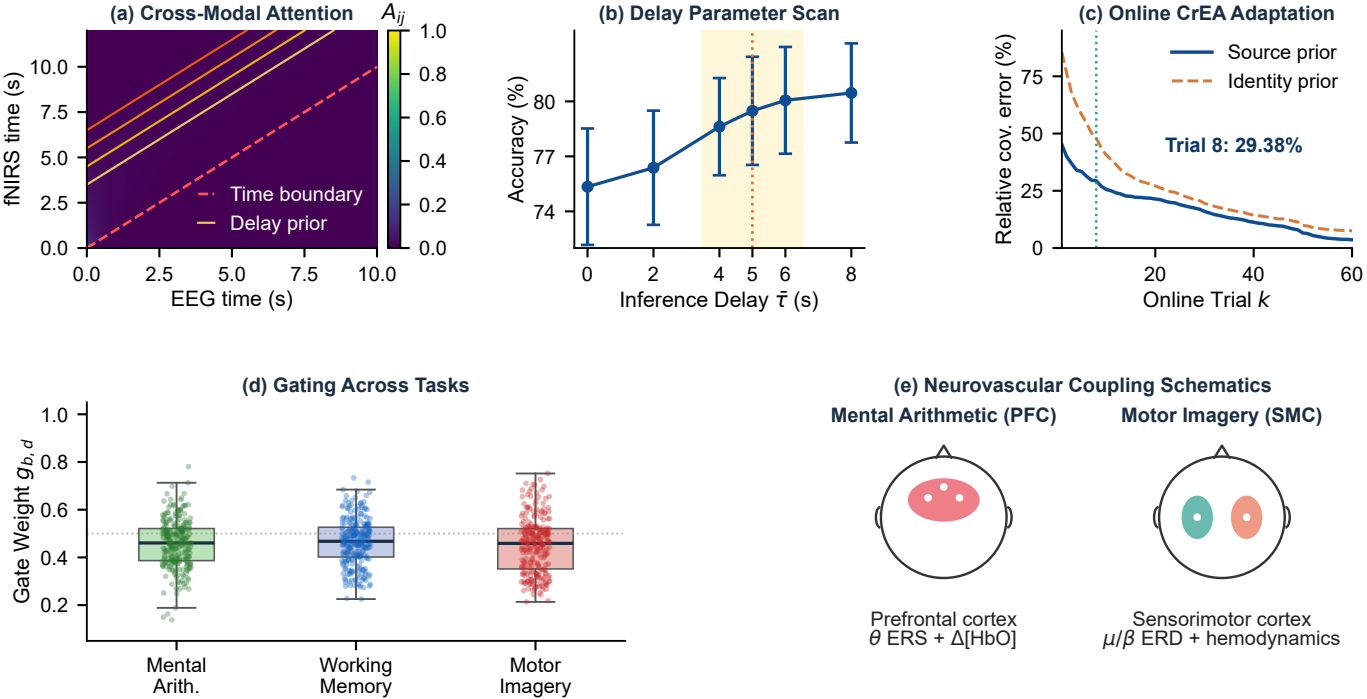

(a) Cross-Modal Attention
A_{ij}
fNIRS time (s)
EEG time (s)
Time boundary
Delay prior
(b) Delay Parameter Scan
Accuracy (%)
Inference Delay $\bar{\tau}$ (s)
(c) Online CrEA Adaptation
Relative cov. error (%)
Online Trial k
Source prior
Identity prior
Trial 8: 29.38%
(d) Gating Across Tasks
Gate Weight $g_{b,d}$
Mental Arith.
Working Memory
Motor Imagery
(e) Neurovascular Coupling Schematics
Mental Arithmetic (PFC)
Motor Imagery (SMC)
Prefrontal cortex
θ ERS + Δ[HbO]
Sensorimotor cortex
μ/β ERD + hemodynamics